

- 7 (a) Fig. 7.1 is a transmission electron micrograph of part of a chloroplast of a leaf cell from maize.

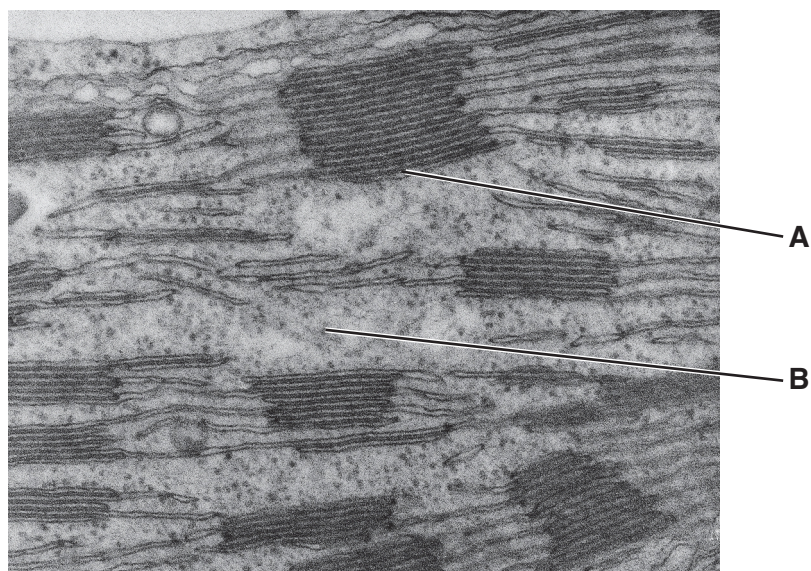


Fig. 7.1

Table 7.1 shows some substrates and products involved in photosynthesis.

Use letter **A** or letter **B** from Fig. 7.1 to complete Table 7.1 to show the location where the substrates or products are used or produced.

Table 7.1

substrate or product	location
oxygen produced	
carbon dioxide used	
reduced NADP used	
ATP produced	
hexose produced	

[3]

- (b) Chloroplasts isolated from leaf palisade cells can still function if they are suspended in a buffer solution. The buffer solution has the same water potential as the chloroplasts.

The dye DCPIP is a hydrogen acceptor that changes colour from blue to colourless when it becomes reduced.

Three test tubes were set up as shown in Table 7.2 and left for 20 minutes to allow any colour change to occur. The results are also shown in Table 7.2.

Table 7.2

test-tube	contents	conditions	colour change
1	buffer solution + DCPIP	light	no
2	chloroplast suspension + DCPIP	light	yes
3	chloroplast suspension + DCPIP	dark	no

(i) Explain the results for test-tube 2.

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[2]

(ii) Test-tube 1 is a control tube. Explain why test-tube 1 was included in the investigation.

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[1]

(iii) Suggest **and** explain what would happen to the chloroplasts if they were suspended in distilled water.

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[2]

(c) The rate of photosynthesis in green plants can be limited by factors such as light intensity, temperature and carbon dioxide concentration.

State which factor would have **no** effect on the reducing ability of a chloroplast suspension. Give a reason for your answer.

factor

reason

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[2]